

Name: _____

SIMULTANEOUS EQUATIONS, (SOLVING BY ELIMINATION)

Date: _____

LO: I can solve simultaneous linear equations by adding or subtracting to eliminate a variable.

The elimination method

The **elimination method** removes one variable by adding or subtracting the equations. We make the **coefficients** of one variable equal (by multiplying), then add or subtract.

Strategy: Same signs = subtract. Different signs = add. Then solve for the remaining variable and substitute back.

Quick examples

●	$2x + y = 7$ and $x + y = 4$: subtract to get $x = 3$	same sign y, subtract
●	$3x + 2y = 12$ and $3x - y = 3$: subtract to eliminate x	same coefficient of x
●	May need to multiply one or both equations first	to match coefficients
●	Always add the equations	sometimes subtract

Key idea

Make coefficients of one variable equal, then add (if different signs) or subtract (if same signs) to eliminate it.

$$2x + 3y = 12 \text{ and } 2x - y = 4:$$

$$\text{subtract } 4y = 8$$

Same coefficient of x (both 2x): subtract equations

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - simple subtraction

$$\text{Solve } 2x + y = 7 \text{ and } x + y = 4$$

Subtract equation 2 from equation

$$1: (2x+y) - (x+y) = 7 - 4$$

$$x = 3$$

$$\text{Substitute: } 3 + y = 4, \text{ so } y = 1$$

$$x = 3, y = 1$$

Both equations have +y, so subtract to eliminate y.

Example 2 - simple addition

$$\text{Solve } 3x + 2y = 16 \text{ and } x - 2y = 0$$

Add the equations: $(3x+2y) + (x-2y)$

$$= 16 + 0$$

$$4x = 16, \text{ so } x = 4$$

$$\text{Substitute: } 4 - 2y = 0, \text{ so } y = 2$$

$$x = 4, y = 2$$

+2y and -2y: add to eliminate y.

Example 3 - multiply one equation

$$\text{Solve } 2x + 3y = 13 \text{ and } x + y = 5$$

Multiply eq 2 by 2: $2x + 2y = 10$

Subtract from eq 1: $(2x+3y) - (2x+2y)$

$$= 13 - 10$$

$$y = 3; \text{ substitute: } x + 3 = 5, x = 2$$

$$x = 2, y = 3$$

Multiply to match the coefficient of x, then subtract.

Example 4 - multiply both equations

$$\text{Solve } 3x + 2y = 12 \text{ and } 5x + 3y = 19$$

Multiply eq 1 by 3: $9x + 6y = 36$.

Multiply eq 2 by 2: $10x + 6y = 38$

Subtract: $10x - 9x = 38 - 36$, so $x = 2$

$$\text{Substitute: } 6 + 2y = 12, y = 3$$

$$x = 2, y = 3$$

Multiply both to make one coefficient the same, then subtract.

YOUR TURN – PRACTICE (deliberate practice)

1. Direct elimination

Solve by adding or subtracting directly.

- a) $x + y = 6$ and $x - y = 2$
- b) $2x + y = 9$ and $2x - y = 1$
- c) $3x + 2y = 14$ and $x + 2y = 10$
- d) $5x + y = 17$ and $3x + y = 11$

2. Multiply one equation

Multiply one equation first, then eliminate.

- a) $2x + y = 8$ and $x + 3y = 9$
- b) $3x + 2y = 16$ and $x + y = 6$
- c) $4x + y = 14$ and $2x + 3y = 12$
- d) $x + 2y = 7$ and $3x + y = 11$

3. Multiply both equations

Multiply both equations to match a coefficient.

- a) $2x + 3y = 13$ and $3x + 2y = 12$
- b) $3x + 4y = 18$ and $5x + 2y = 16$
- c) $4x + 3y = 23$ and $3x + 5y = 24$
- d) $2x + 5y = 19$ and $3x + 4y = 18$

4. Negative coefficients

Solve each pair.

- a) $2x - y = 5$ and $x + y = 7$
- b) $3x - 2y = 5$ and $x + 2y = 7$
- c) $4x - 3y = 1$ and $2x + 3y = 17$
- d) $5x - y = 13$ and $2x + 3y = 12$

5. Mixed problems

Solve each pair of simultaneous equations.

- a) $x + y = 8$ and $2x + 3y = 21$
- b) $3x + 2y = 19$ and $x - y = 2$
- c) $4x + y = 10$ and $x + 4y = 10$
- d) $2x + 5y = 1$ and $3x - 2y = -12$
- e) $x + 2y = 10$ and $3x - y = 9$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Same signs: subtract equations. Different signs: add equations ☐

Correct answer: _____

Reason: _____

b) You can only use elimination if coefficients already match ☐

Correct answer: _____

Reason: _____

c) After finding one variable, substitute back to find the other ☐

Correct answer: _____

Reason: _____

d) Always add the equations together ☐

Correct answer: _____

Reason: _____

e) You can multiply one or both equations to create matching coefficients ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Solve: $3x + 4y = -2$ and $5x - 3y = 16$. Check your answer satisfies both equations.

Expression: _____

Simplified: _____

b) Two numbers have a sum of **44** and a difference of **12**. Find both numbers. Then find two numbers whose sum is 100 and where one is three times the other.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

